

VIKRAMADITYA SINGH

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Profile

Software Engineer with ≈ 4 years of experience specializing in system design, algorithms, operating systems, and database management. Skilled in developing scalable, efficient software solutions with a strong foundation in data structures and performance optimization. Adept at collaborating with cross-functional teams, communicating complex technical concepts clearly, and adapting to evolving project requirements. Demonstrates a deep technical understanding of full-stack development and cloud platforms, maintaining a strong commitment to high standards of code quality, system reliability, and continuous improvement.

Experience

CISCO

Bangalore, India

Software Engineer III

August 2021 - present

- Designed, developed, and scaled the backend for the Policy Analyser and Optimiser microservice, enabling the analysis of all Access Policies across FMCs for all tenants on CDO.
- Implemented core analysis logic for rule conflict, object overlap, expired rules, mergeable rules, and never-hit rules detection.
- Created an Access Policy converter microservice, making the Policy Analyser and Optimiser (PAO) version-agnostic, which significantly expanded the customer base.
- Built and maintained CI/CD pipelines in Jenkins, integrating alerting and metrics collection with Datadog.
- Developed a Remediation service to address anomalies identified by the Policy Analyser and Optimiser.
- Created a pub/sub framework for consumers like AI Assistants & AIOPs to subscribe to analysis events.
- Enhanced NAP & IPS policy snapshot creation, cutting operation time by 70% and reducing FMC device configuration deployment time.
- Designed and implemented Zero Trust policy on FMC, including full backend development (API, global search, database, services, validations, reporting, audit, telemetry).
- Developed changes for Zero Trust policy deployment, adding Snort3 configuration creation and enhancing Lina CLI generation with new commands.
- Implemented auto-enrollment of certificates within the Zero Trust policy, significantly enhancing device integration and user experience.
- Improved FMC device listing and management page performance, reducing load time by 96%.
- Added API support for Snort3 rule querying based on the MITRE framework.
- Implemented changes to display rule movement in the audit log, enhancing the user experience.
- Added Elephant Flow Detection to the Access Policy, including backend and deployment changes for Lina and Snort3.
- Integrated PDF reporting, audit logs, delta preview, and telemetry for Elephant Flow Detection & Threat Detection settings.
- Implemented dynamic warning framework for FMC device upgrade flows using React.

SECURONIX

Pune, India

SDE Intern

August 2020 - April 2021

- Developed email microservice and automated incident report generation using Spring Framework and Jasper Reports.
- Created scripts to automate various task.
- Automated report creation for analysis from the data for CTA team.

CISCO

Bangalore, India

SDE Intern

May 2020 - July 2020

- Developed a gRPC detector for Snort 2.9.16 and integrated it with FTD to enable alert generation.
- Worked on the preprocessor framework for scanning the message content transferred via. gRPC for malicious payloads.

Publication

Inception Time Model for Structural Damage Detection Using Vibration Measurements

March 2024

In book Fourth Congress on Intelligent Systems (pp.103-122), Springer, Singapore, 31 March 2024

DOI: 10.1007/978-981-99-9040-5_7

Projects

Domain Generating Algorithm Detection | *Python, Keras, Pandas, Google Collab* **code**

- Detected algorithmically generated domains with 96% accuracy and classified them by DGA variant with 87% accuracy, using clustering to identify unknown DGA variants.

Sort Visualizer | *NextJs, Vercel* **demo** | **code**

- Developed a dynamic web application that visually demonstrates the functionality of various sorting algorithms like Quick Sort, Merge Sort, Bubble Sort, and more.

Portfolio Site | *NextJs, Vercel, Slack* **demo**

- Developed the frontend using Next.js and Tailwind CSS, deployed on Vercel, and integrated with Slack for messaging.
- Utilized Ummami for generating site analytics.

Tech Blog : NeuralCook | *NextJs, Vercel, AWS Lambda , FireStore, MailGun, Github Action* **demo** | **code**

- Developed the frontend using Next.js and tailwind leveraged AWS Lambda and Firestore for the backend.
- Automated blog publication using GitHub Actions, utilizing GitHub as a CMS.
- Integrated with Slack for notifications and Mailgun for sending blog email alerts.

Achievements

Smart India Hackathon **2019**

Finalist for hardware edition

- Developed a mobile-based irrigation assistant app using Flutter for the frontend and Firebase for backend integration.
- Deployed a LoRaWAN network to extend range and overcome connectivity limitations.

Awards

CISCO

- Connect Everything x4
- Going Above and Beyond x2
- Living Cisco's Principles x1
- Driving the Highest Level of Achievement x1

Education

Indian Institute of Information Technology, Guwahati **2017-2021**

Bachelor of Technology in Computer Science Engineering

Cum. GPA: 8.48

Modern Delhi Public School, Faridabad **2015 - 2016**

12th Science from CBSE Board

87%

Dynasty International School, Faridabad **2013-2014**

10th from CBSE Board

Cum. GPA: 9.8

Technical Skills

Languages: Java, SQL, Javascript, Python, C, Perl

Frameworks: ReactJs, NextJs, Spring, Spring Boot, Hibernate

Technologies: Docker, Jenkins, Git, Perforce, Terraform, Lucene

Databases: MariaDB, MongoDB, Redis

Cloud: AWS

Developer Tools: IntelliJ IDEA, Postman, Data Dog, Draw.io, LocalStack

Relevant Coursework

- Data Structure & Algorithms
- Computer Network & Security
- Operating System
- RDBMS
- OOP & Design Patterns
- Software Engineering
- Deep Learning